Machine Learning

Assignment 1

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Q1. Ridge, Lasso and the duplicated-feature trap

(1) Derive the ridge solution, prove invertibility, and give the augmented least-squares problem.

Solution.

(a) Derivation of the Ridge Solution

Given the objective function

$$F(w) = \|y - Xw\|_2^2 + \lambda\|w\|_2^2,$$

its gradient is

$$\nabla F(w) = -2X^\top(y - Xw) + 2\lambda w.$$

Setting the gradient equal to zero,

$$-2X^\top(y - Xw) + 2\lambda w = 0.$$

This simplifies to

$$(X^\top X + \lambda I)\widehat{w} = X^\top y.$$

Hence, the closed-form ridge regression solution is

$$\boxed{\widehat{w} = (X^\top X + \lambda I)^{-1}X^\top y.}$$

(b) Proof that $X^\top X + \lambda I$ is Invertible

For any nonzero vector u,

$$u^\top(X^\top X + \lambda I)u = u^\top X^\top Xu + \lambda u^\top u.$$

Since

$$u^\top X^\top Xu = \|Xu\|_2^2 \geq 0$$

and

$$\lambda > 0, \qquad u^\top u = \|u\|_2^2 > 0,$$

we have

$$u^\top(X^\top X + \lambda I)u = \|Xu\|_2^2 + \lambda\|u\|_2^2 > 0.$$

Therefore,

$$X^\top X + \lambda I$$

is positive definite and hence invertible, even when X is rank-deficient.

(c) Ridge as an Augmented Least-Squares Problem
Observe that

$$\|y - Xw\|_2^2 + \lambda\|w\|_2^2 = \left\| \begin{bmatrix} y \\ 0_d \end{bmatrix} - \begin{bmatrix} X \\ \sqrt{\lambda} I_d \end{bmatrix} w \right\|_2^2.$$

Thus, ridge regression is exactly ordinary least squares on the augmented dataset

$$X_{\text{aug}} = \begin{bmatrix} X \\ \sqrt{\lambda} I_d \end{bmatrix}, \qquad y_{\text{aug}} = \begin{bmatrix} y \\ 0_d \end{bmatrix}.$$

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(2) Effect of duplicating a feature in ridge regression.

Solution.

(a) Ridge assigns identical weights to all duplicated features
Suppose the first feature x_1 is duplicated m times, giving the new design matrix

$$X' = [x_1, x_1, \ldots, x_1, X_{-1}].$$

Let the corresponding weights be

$$w = (w_1, w_2, \ldots, w_m, w_{m+1}, \ldots).$$

The ridge objective is

$$J(w) = \|y - X'w\|_2^2 + \lambda\|w\|_2^2.$$

Since all m columns are identical, exchanging any two weights does not change the prediction

$$X'w.$$

Also,

$$\|w\|_2^2 = w_1^2 + w_2^2 + \cdots + w_m^2 + \cdots$$

is unchanged under any permutation of the duplicated weights.
Thus the objective function is symmetric with respect to $w_1, \ldots, w_m$.
Since ridge regression is strictly convex, the minimizer is unique. The only unique solution that is invariant under every permutation is

$$\boxed{w_1 = w_2 = \cdots = w_m.}$$

—

(b) Total effect of the duplicated feature
Let

$$\beta = \sum_{j=1}^{m} w_j.$$

Since all duplicated weights are equal,

$$w_j = \frac{\beta}{m}, \qquad j = 1, \ldots, m.$$

The prediction becomes

$$X'w = x_1 \sum_{j=1}^{m} w_j + X_{-1}w_{-1} = \beta x_1 + X_{-1}w_{-1}.$$

The penalty term is

$$\lambda \sum_{j=1}^{m} w_j^2 = \lambda m \left(\frac{\beta}{m}\right)^2 = \frac{\lambda}{m}\beta^2.$$

Therefore the ridge objective becomes

$$\|y - \beta x_1 - X_{-1}w_{-1}\|_2^2 + \frac{\lambda}{m}\beta^2 + \lambda\|w_{-1}\|_2^2.$$

This is exactly the ridge regression objective on the original design matrix, except that the penalty on the first coefficient is

$$\boxed{\frac{\lambda}{m}}$$

instead of

$$\lambda.$$

—

(c) Conclusion

Duplicating a feature does not change its total contribution

$$\beta = \sum_{j=1}^{m} w_j,$$

but it distributes the coefficient equally among the copies.

The effective regularization on that feature decreases from

$$\lambda$$

to

$$\boxed{\frac{\lambda}{m}}.$$

Hence, duplicating a feature weakens its regularization, allowing the duplicated feature to receive a larger overall coefficient than a non-duplicated feature.

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(3) Lasso and Elastic Net with duplicated features.

Solution.

(a) Lasso solution is not unique

Consider the Lasso objective

$$J(w) = \|y - X'w\|_2^2 + \lambda\|w\|_1,$$

where

$$X' = [x_1, x_1, \ldots, x_1, X_{-1}].$$

Let

$$\beta = \sum_{j=1}^{m} w_j.$$

The prediction depends only on the sum

$$X'w = \beta x_1 + X_{-1}w_{-1}.$$

The Lasso penalty for the duplicated coefficients is

$$\sum_{j=1}^{m} |w_j|.$$

If all duplicated weights have the same sign,

$$\sum_{j=1}^{m} |w_j| = \left| \sum_{j=1}^{m} w_j \right| = |\beta|.$$

Hence the objective depends only on

$$\beta,$$

not on how it is distributed among the duplicated features.
Therefore there are infinitely many optimal solutions.
Thus,

$$\boxed{\text{The Lasso solution is generally not unique.}}$$

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(b) Characterization of all optimal weight splits
Every optimal solution satisfies

$$\sum_{j=1}^{m} w_j = \beta,$$

together with

$$\sum_{j=1}^{m} |w_j| = |\beta|.$$

This occurs exactly when all duplicated coefficients have the same sign (or some are zero).
Hence the complete set of optimal solutions is

$$\boxed{\left\{ (w_1, \ldots, w_m) : \sum_{j=1}^{m} w_j = \beta,\ \sum_{j=1}^{m} |w_j| = |\beta| \right\}.}$$

Thus the total coefficient is fixed, but it may be split among the duplicated variables in infinitely many ways.

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(c) Elastic Net forces equal weights (Grouping Effect)
The Elastic Net objective is

$$J(w) = \|y - X'w\|_2^2 + \lambda_1 \|w\|_1 + \lambda_2 \|w\|_2^2, \qquad \lambda_2 > 0.$$

Again let

$$\beta = \sum_{j=1}^{m} w_j.$$

The prediction and Lasso penalty depend only on

$$\beta,$$

while the quadratic penalty is

$$\lambda_2 \sum_{j=1}^{m} w_j^2.$$

For a fixed value of

$$\beta,$$

the quantity

$$\sum_{j=1}^{m} w_j^2$$

is minimized when all duplicated coefficients are equal.
Using Jensen's inequality (or Cauchy–Schwarz),

$$\sum_{j=1}^{m} w_j^2 \geq \frac{\left(\sum_{j=1}^{m} w_j\right)^2}{m} = \frac{\beta^2}{m},$$

with equality if and only if

$$w_1 = w_2 = \cdots = w_m.$$

Therefore the Elastic Net has the unique solution

$$\boxed{w_1 = w_2 = \cdots = w_m = \frac{\beta}{m}.}$$

This phenomenon is called the **grouping effect**: highly correlated or duplicated features receive nearly identical coefficients.

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(4) Why ranking features by $|\hat{w}_j|$ from Lasso can be misleading.

Solution.

From Part (2), duplicating a feature splits its total effect among the copies. The overall contribution is measured by

$$\beta = \sum_{j=1}^{m} w_j,$$

rather than by any individual coefficient.

From Part (3), the Lasso solution is not unique. The same total coefficient β can be distributed among duplicated features in infinitely many ways while producing the same prediction and objective value.

Therefore, a duplicated feature may receive a very small (or even zero) coefficient even though the combined effect of all its copies is large.

Hence, ranking features solely by $|\hat{w}_j|$ can incorrectly label an important feature as "unimportant." Instead, the combined coefficient of duplicated or highly correlated features should be considered.

Q2. The squared-hinge SVM is secretly a hard-margin SVM

(1) Prove that adding $\xi_i \geq 0$ does not change the optimum, and explain why this is not true for the standard L1-SVM.

Solution.

(a) The constraint $\xi_i \geq 0$ is redundant for the L2-SVM

The L2-SVM optimization problem is

$$\min_{w,b,\xi} \frac{1}{2}\|w\|^2 + \frac{C}{2}\sum_{i=1}^{n} \xi_i^2$$

subject to

$$y_i \left(w^\top \phi(x_i) + b \right) \geq 1 - \xi_i, \qquad i = 1, \ldots, n.$$

Assume an optimal solution has some

$$\xi_i < 0.$$

Then

$$1 - \xi_i > 1,$$

so the constraint becomes

$$y_i(w^\top \phi(x_i) + b) \geq 1 - \xi_i > 1.$$

Hence the same point also satisfies

$$y_i(w^\top \phi(x_i) + b) \geq 1,$$

which means replacing

$$\xi_i$$

by

$$0$$

still satisfies the constraint.

Since

$$0^2 < \xi_i^2,$$

the objective value decreases because

$$\frac{C}{2}(0)^2 < \frac{C}{2}\xi_i^2.$$

This contradicts the assumption that the original solution was optimal.

Therefore,

$$\boxed{\xi_i \geq 0}$$

must hold automatically at the optimum.
Hence adding the constraint

$$\xi_i \geq 0$$

does not change the solution.

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(b) Why this is not true for the standard L1-SVM
The L1-SVM objective is

$$\min_{w,b,\xi} \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n} \xi_i.$$

If the constraint

$$\xi_i \geq 0$$

is removed, then negative slack values are allowed.
Since the objective contains

$$C\xi_i$$

instead of

$$C\xi_i^2,$$

making

$$\xi_i$$

more negative always decreases the objective.
At the same time, the constraint

$$y_i(w^\top \phi(x_i) + b) \geq 1 - \xi_i$$

can still be satisfied by increasing the margin.
Thus the optimization can continue decreasing the objective by choosing increasingly negative slack values.
Therefore the problem is not well-defined without the constraint

$$\xi_i \geq 0.$$

Hence,

$$\boxed{\text{For the L1-SVM, the constraint } \xi_i \geq 0 \text{ is essential and cannot be removed.}}$$

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(2) Derive the Lagrangian dual. Show that it is the hard-margin dual with kernel matrix $K + \frac{1}{C}I$, and that the box constraint $0 \leq \alpha_i \leq C$ disappears.

Solution.

(a) Form the Lagrangian
Introduce Lagrange multipliers

$$\alpha_i \geq 0,$$

for the constraints

$$y_i(w^\top \phi(x_i) + b) \geq 1 - \xi_i.$$

The Lagrangian is

$$L(w, b, \xi, \alpha) = \frac{1}{2}\|w\|^2 + \frac{C}{2}\sum_{i=1}^{n} \xi_i^2 - \sum_{i=1}^{n} \alpha_i \left(y_i(w^\top \phi(x_i) + b) - 1 + \xi_i \right).$$

—

(b) Stationarity Conditions

Taking derivatives with respect to the primal variables,

$$\frac{\partial L}{\partial w} = w - \sum_{i=1}^{n} \alpha_i y_i \phi(x_i) = 0,$$

which gives

$$\boxed{w = \sum_{i=1}^{n} \alpha_i y_i \phi(x_i).}$$

Differentiating with respect to b,

$$\frac{\partial L}{\partial b} = -\sum_{i=1}^{n} \alpha_i y_i = 0,$$

hence

$$\boxed{\sum_{i=1}^{n} \alpha_i y_i = 0.}$$

Differentiating with respect to ξ_i,

$$\frac{\partial L}{\partial \xi_i} = C\xi_i - \alpha_i = 0,$$

so

$$\boxed{\xi_i = \frac{\alpha_i}{C}.}$$

—

(c) Dual Objective

Substituting these results into the Lagrangian,

$$g(\alpha) = \sum_{i=1}^{n} \alpha_i - \frac{1}{2}\sum_{i,j} \alpha_i \alpha_j y_i y_j K(x_i, x_j) - \frac{1}{2C}\sum_{i=1}^{n} \alpha_i^2.$$

Since

$$\frac{1}{2C}\sum_{i} \alpha_i^2 = \frac{1}{2}\sum_{i,j} \alpha_i \alpha_j \frac{\delta_{ij}}{C},$$

the dual becomes

$$\boxed{g(\alpha) = \sum_i \alpha_i - \frac{1}{2}\sum_{i,j} \alpha_i \alpha_j y_i y_j \left(K(x_i, x_j) + \frac{\delta_{ij}}{C} \right).}$$

Thus the dual is exactly the hard-margin SVM dual with kernel matrix

$$\boxed{K + \frac{1}{C} I.}$$

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(d) Absence of the Box Constraint

The dual optimization problem is

$$\max_{\alpha} \; g(\alpha)$$

subject to

$$\alpha_i \geq 0,$$

and

$$\sum_i \alpha_i y_i = 0.$$

Unlike the L1-SVM,

$$\boxed{\text{there is no upper bound } \alpha_i \leq C.}$$

This is because the quadratic penalty

$$\frac{C}{2}\xi_i^2$$

produces the relation

$$\xi_i = \frac{\alpha_i}{C},$$

instead of restricting α_i to the interval $[0, C]$.

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(3) From the KKT conditions, derive the relation between ξ_i and α_i. Classify which training points have $\alpha_i > 0$ and compare with the L1-SVM.

Solution.

(a) Relation between ξ_i and α_i

From the stationarity condition of the Lagrangian,

$$\frac{\partial L}{\partial \xi_i} = 0,$$

we obtain

$$C\xi_i - \alpha_i = 0.$$

Hence,

$$\boxed{\xi_i = \frac{\alpha_i}{C}.}$$

Thus, every slack variable is directly proportional to its corresponding dual variable.

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(b) Classification of Training Points

The KKT complementary slackness condition is

$$\alpha_i \Big(y_i(w^\top \phi(x_i) + b) - 1 + \xi_i \Big) = 0.$$

There are two cases.

Case 1: $\alpha_i = 0$

Then

$$\xi_i = 0,$$

and the point satisfies

$$y_i(w^\top \phi(x_i) + b) \geq 1.$$

These points lie outside the margin and are not support vectors.

—

Case 2: $\alpha_i > 0$

Then

$$y_i(w^\top \phi(x_i) + b) = 1 - \xi_i.$$

Since

$$\xi_i = \frac{\alpha_i}{C},$$

we obtain

$$y_i(w^\top \phi(x_i) + b) = 1 - \frac{\alpha_i}{C}.$$

These points lie exactly on the margin or inside the margin and are support vectors.
Hence,

$$\boxed{\alpha_i > 0 \iff \text{support vector.}}$$

—

(c) Comparison with the L1-SVM

For the L1-SVM, the dual variables satisfy

$$0 \leq \alpha_i \leq C.$$

Thus,

- Points outside the margin have

$$\alpha_i = 0.$$

- Points exactly on the margin satisfy

$$0 < \alpha_i < C.$$

- Misclassified or margin-violating points often satisfy

$$\alpha_i = C,$$

because of the upper bound.

For the L2-SVM,

$$\boxed{\alpha_i \geq 0}$$

with no upper bound.
Since

$$\xi_i = \frac{\alpha_i}{C},$$

larger violations simply produce larger values of

$$\alpha_i,$$

instead of clipping them at

$$C.$$

Therefore,

Unlike the L1-SVM, the L2-SVM has no upper bound on α_i, and support vectors are not limited by C.

---

(3) From the KKT conditions, derive the relation between ξ_i and α_i. Classify which training points have $\alpha_i > 0$ and compare with the L1-SVM.

Solution.

(a) Relation between ξ_i and α_i
From the stationarity condition of the Lagrangian,

$$\frac{\partial L}{\partial \xi_i} = 0,$$

we obtain

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These points lie outside the margin and are not support vectors.

—

Case 2: $\alpha_i > 0$
Then

$$y_i(w^\top \phi(x_i) + b) = 1 - \xi_i.$$

Since

$$\xi_i = \frac{\alpha_i}{C},$$

we obtain

$$y_i(w^\top \phi(x_i) + b) = 1 - \frac{\alpha_i}{C}.$$

These points lie exactly on the margin or inside the margin and are support vectors.
Hence,

$$\boxed{\alpha_i > 0 \iff \text{support vector.}}$$

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- Points outside the margin have

$$\alpha_i = 0.$$

- Points exactly on the margin satisfy

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- Misclassified or margin-violating points often satisfy

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because of the upper bound.
For the L2-SVM,

$$\boxed{\alpha_i \geq 0}$$

with no upper bound.
Since

$$\xi_i = \frac{\alpha_i}{C},$$

larger violations simply produce larger values of

$$\alpha_i,$$

instead of clipping them at

$$C.$$

Therefore,

Unlike the L1-SVM, the L2-SVM has no upper bound on α_i, and support vectors are not limited by C.

Q3. Naive Bayes counts the same evidence twice

(1) Show that when the variance s_j^2 is shared across the two classes, the Naive Bayes posterior has the logistic form. Give w and b explicitly.

Solution.

(a) Gaussian Naive Bayes Model

For class $y = k$ $(k = 0, 1)$,

$$P(x_j|y = k) = \frac{1}{\sqrt{2\pi s_j^2}} \exp\left(-\frac{(x_j - \mu_{jk})^2}{2s_j^2}\right).$$

Since Naive Bayes assumes conditional independence,

$$P(x|y = k) = \prod_{j=1}^{d} P(x_j|y = k).$$

Using Bayes' rule,

$$P(y = 1|x) = \frac{P(x|y = 1)P(y = 1)}{P(x|y = 1)P(y = 1) + P(x|y = 0)P(y = 0)}.$$

—

(b) Compute the Log-Odds

The posterior log-odds are

$$\log \frac{P(y = 1|x)}{P(y = 0|x)} = \log \frac{P(x|y = 1)\pi}{P(x|y = 0)(1 - \pi)}.$$

Substituting the Gaussian densities,

$$= \log \frac{\pi}{1 - \pi} + \sum_{j=1}^{d} \frac{(x_j - \mu_{j0})^2 - (x_j - \mu_{j1})^2}{2s_j^2}.$$

Expanding,

$$(x_j - \mu_{j0})^2 - (x_j - \mu_{j1})^2 = 2x_j(\mu_{j1} - \mu_{j0}) + \mu_{j0}^2 - \mu_{j1}^2.$$

Hence,

$$\log \frac{P(y = 1|x)}{P(y = 0|x)} = \sum_{j=1}^{d} \frac{\mu_{j1} - \mu_{j0}}{s_j^2} x_j + \log \frac{\pi}{1 - \pi} + \sum_{j=1}^{d} \frac{\mu_{j0}^2 - \mu_{j1}^2}{2s_j^2}.$$

—

(c) Logistic Form

Define

$$\boxed{w_j = \frac{\mu_{j1} - \mu_{j0}}{s_j^2}}$$

and

$$\boxed{b = \log \frac{\pi}{1-\pi} + \sum_{j=1}^{d} \frac{\mu_{j0}^2 - \mu_{j1}^2}{2s_j^2}.}$$

Therefore,

$$\boxed{\log \frac{P(y=1|x)}{P(y=0|x)} = w^\top x + b.}$$

Applying the sigmoid function,

$$\boxed{P(y=1|x) = \sigma(w^\top x + b),}$$

where

$$\sigma(z) = \frac{1}{1+e^{-z}}.$$

Thus, when both classes share the same feature variances, Gaussian Naive Bayes produces exactly the same logistic posterior form as Logistic Regression.

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(2) Show that when the variances are class-specific, s_{jk}^2, the decision boundary is quadratic but contains no cross terms. Explain how this differs from QDA with full covariance matrices.

Solution.

(a) Gaussian Naive Bayes with Class-Specific Variances

Suppose

$$x_j|y=k \sim N(\mu_{jk},\, s_{jk}^2),$$

where each class has its own variance.

The conditional density is

$$P(x|y=k) = \prod_{j=1}^{d} \frac{1}{\sqrt{2\pi s_{jk}^2}} \exp\left(-\frac{(x_j-\mu_{jk})^2}{2s_{jk}^2}\right).$$

The log-posterior ratio becomes

$$\log \frac{P(y=1|x)}{P(y=0|x)} = \log \frac{\pi}{1-\pi} + \log \frac{P(x|1)}{P(x|0)}.$$

—

(b) Decision Boundary

Expanding the Gaussian terms,

$$\log \frac{P(y=1|x)}{P(y=0|x)} = c + \sum_{j=1}^{d} \left(a_j x_j^2 + b_j x_j\right),$$

where

$$a_j = \frac{1}{2s_{j0}^2} - \frac{1}{2s_{j1}^2},$$

and

$$b_j = \frac{\mu_{j1}}{s_{j1}^2} - \frac{\mu_{j0}}{s_{j0}^2},$$

and c contains all constant terms.
Thus the decision boundary is

$$\boxed{c + \sum_{j=1}^{d} \left(a_j x_j^2 + b_j x_j\right) = 0.}$$

Hence the boundary is quadratic.

—

(c) No Cross Terms
Notice that every quadratic term involves only

$$x_j^2.$$

There are no products such as

$$x_i x_j, \qquad i \neq j.$$

Therefore,

The Gaussian Naive Bayes decision boundary contains only x_j^2 terms and no cross terms.

This follows directly from the conditional independence assumption.

—

(d) Comparison with QDA
Quadratic Discriminant Analysis (QDA) uses a full covariance matrix

$$\Sigma_k.$$

The quadratic form is

$$(x - \mu_k)^T \Sigma_k^{-1} (x - \mu_k),$$

which expands to

$$\sum_i a_i x_i^2 + \sum_{i \neq j} b_{ij} x_i x_j.$$

Thus QDA contains interaction terms

$$x_i x_j,$$

whenever the covariance matrix has nonzero off-diagonal elements.
Hence,

Gaussian Naive Bayes: Quadratic boundary with only x_j^2.
QDA: Quadratic boundary containing both x_j^2 and $x_i x_j$ terms.

The absence of cross terms in Naive Bayes is a consequence of assuming conditional independence between features.

(3) Suppose feature x_1 is copied m times, so $x_1 = x_2 = \cdots = x_m$. Show that Naive Bayes multiplies the log-odds contribution by m. If the true posterior is $\sigma(a(x))$, explain why Naive Bayes becomes over-confident as m grows. Then show that Logistic Regression is unaffected because only the sum of the m weights is identified.

Solution.

(a) Naive Bayes with Duplicated Features

The Gaussian Naive Bayes log-odds are

$$\log \frac{P(y=1|x)}{P(y=0|x)} = b + \sum_{j=1}^{d} w_j x_j.$$

Suppose feature x_1 is duplicated m times,

$$x_1 = x_2 = \cdots = x_m.$$

Since each copy contributes the same term,

$$w_1 x_1,$$

their total contribution becomes

$$\boxed{m w_1 x_1.}$$

Therefore, the log-odds become

$$\boxed{b + m w_1 x_1 + \sum_{j>m} w_j x_j.}$$

Hence, Naive Bayes counts the same evidence m times.

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(b) Over-Confident Probabilities

Assume the true posterior probability is

$$P(y=1|x) = \sigma(a(x)).$$

After duplicating the feature,

$$P_{\text{NB}}(y=1|x) = \sigma(m a(x)).$$

As

$$m \to \infty,$$

the sigmoid function becomes increasingly steep,

$$\sigma(m a(x)) \to \begin{cases} 1, & a(x) > 0, \\ 0, & a(x) < 0. \end{cases}$$

Thus Naive Bayes predicts probabilities very close to 0 or 1.
Therefore,

Duplicating a feature makes Naive Bayes increasingly over-confident because the same evidence is counted re

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(c) Logistic Regression

Logistic Regression predicts

$$P(y = 1|x) = \sigma(w^\top x + b).$$

If x_1 is duplicated,

$$w_1x_1 + w_2x_1 + \cdots + w_mx_1 = (w_1 + w_2 + \cdots + w_m)x_1.$$

Hence only the total weight

$$\boxed{\beta = \sum_{j=1}^{m} w_j}$$

affects the prediction.
The model becomes

$$P(y = 1|x) = \sigma(\beta x_1 + \cdots + b),$$

which is identical to using a single copy of the feature.
Therefore,

Logistic Regression predictions are unaffected by feature duplication because only the sum of the duplicated

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(4) Ng and Jordan (2002) showed that Naive Bayes reaches its asymptotic error with $O(\log d)$ samples, while Logistic Regression needs $O(d)$ samples, yet Logistic Regression's asymptotic error is never worse. Construct two simple 2-D data distributions: one where both reach the same asymptotic error, and one where Logistic Regression is strictly better in the limit. Justify each using parts (1) and (3).

Solution.

(a) Distribution where both methods have the same asymptotic error
Consider two independent Gaussian features.
For class $y = 0$,

$$x_1, x_2 \sim N(0, 1),$$

and for class $y = 1$,

$$x_1, x_2 \sim N(2, 1).$$

The covariance matrix is

$$\Sigma = I.$$

Thus,
- the features are conditionally independent, - both classes have equal variances.
These satisfy exactly the assumptions of Gaussian Naive Bayes.
From part (1), the posterior becomes

$$P(y = 1|x) = \sigma(w^\top x + b),$$

which is also the Logistic Regression model.
Therefore,

Both Naive Bayes and Logistic Regression converge to the same optimal classifier and have the same asympt

—

(b) Distribution where Logistic Regression is better

Consider correlated Gaussian features.

For class $y = 0$,

$$x \sim N\left(\begin{bmatrix}0\\0\end{bmatrix}, \begin{bmatrix}1 & 0.9\\0.9 & 1\end{bmatrix}\right),$$

and for class $y = 1$,

$$x \sim N\left(\begin{bmatrix}2\\2\end{bmatrix}, \begin{bmatrix}1 & 0.9\\0.9 & 1\end{bmatrix}\right).$$

The features are highly correlated, violating the conditional independence assumption of Naive Bayes.

From part (3), Naive Bayes counts correlated evidence multiple times, producing exaggerated log-odds and over-confident predictions.

Logistic Regression does not assume feature independence. Its prediction depends only on

$$w^\top x + b,$$

and correlated features simply share the total weight.

Hence,

Logistic Regression achieves a lower asymptotic classification error than Naive Bayes.

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(c) Conclusion

If the Naive Bayes assumptions hold exactly, both methods have the same asymptotic error.

When the independence assumption is violated, Logistic Regression performs strictly better because it does not double-count correlated evidence.

Q4. Activation functions: equivalences, gradients and a hidden minimum

(1) Prove that $\tanh(t) = 2\sigma(2t) - 1$. Rewrite a one-hidden-layer tanh network as an equivalent sigmoid network. Extend the argument to a network with two tanh hidden layers.

Solution.

(a) Proof that $\tanh(t) = 2\sigma(2t) - 1$

The sigmoid function is

$$\sigma(t) = \frac{1}{1 + e^{-t}}.$$

Therefore,

$$2\sigma(2t) - 1 = \frac{2}{1 + e^{-2t}} - 1.$$

Simplifying,

$$= \frac{2 - (1 + e^{-2t})}{1 + e^{-2t}} = \frac{1 - e^{-2t}}{1 + e^{-2t}}.$$

Multiplying numerator and denominator by e^{2t},

$$= \frac{e^{2t} - 1}{e^{2t} + 1}.$$

Since

$$\tanh(t) = \frac{e^t - e^{-t}}{e^t + e^{-t}} = \frac{e^{2t} - 1}{e^{2t} + 1},$$

we obtain

$$\boxed{\tanh(t) = 2\sigma(2t) - 1.}$$

—

(b) One-Hidden-Layer Network

Consider the tanh network

$$f(x) = V \tanh(Wx + b) + c.$$

Using

$$\tanh(z) = 2\sigma(2z) - 1,$$

we obtain

$$f(x) = V\,(2\sigma(2Wx + 2b) - 1) + c.$$

Expanding,

$$= 2V\sigma(2Wx + 2b) - V + c.$$

Hence

$$\boxed{f(x) = V'\sigma(W'x + b') + c',}$$

where

$$\boxed{V' = 2V, \qquad W' = 2W, \qquad b' = 2b, \qquad c' = c - V.}$$

Thus every one-hidden-layer tanh network can be written exactly as a sigmoid network.

—

(c) Two-Hidden-Layer Network

Consider

$$f(x) = V \tanh(W_2 \tanh(W_1 x + b_1) + b_2) + c.$$

Replace the first tanh by

$$2\sigma(2z) - 1.$$

The constant -1 becomes part of the affine transformation of the second layer,

$$W_2(2\sigma(\cdot) - 1) + b_2 = 2W_2\sigma(\cdot) + (b_2 - W_2).$$

Replacing the second tanh similarly gives another constant shift, which is absorbed into the output bias.

Therefore,

In a multi-layer network, each -1 shift is absorbed by the bias of the following affine layer.

For the final hidden layer, the shift is absorbed into the output bias.

---

(2) Show that $\sigma'(t) \leq \frac{1}{4}$ and $\tanh'(t) \leq 1$. Hence derive the gradient bound

$$\left\| \frac{\partial L}{\partial h_1} \right\| \leq \left(\max_t |\phi'(t)| \right)^{L-1} \prod_{l=2}^{L} \|W_l\|_2 \left\| \frac{\partial L}{\partial h_L} \right\|.$$

For sigmoid, determine the spectral norms needed to avoid vanishing gradients and explain why making them very large causes problems.

Solution.

(a) Derivative of the Sigmoid Function

The sigmoid function is

$$\sigma(t) = \frac{1}{1 + e^{-t}}.$$

Differentiating,

$$\sigma'(t) = \sigma(t)\big(1 - \sigma(t)\big).$$

Since

$$0 < \sigma(t) < 1,$$

the product

$$\sigma(t)(1 - \sigma(t))$$

is maximized when

$$\sigma(t) = \frac{1}{2}.$$

Hence,

$$\boxed{\sigma'(t) \leq \frac{1}{4}.}$$

—

(b) Derivative of the Tanh Function

The derivative of the hyperbolic tangent is

$$\tanh'(t) = 1 - \tanh^2(t).$$

Since

$$-1 \leq \tanh(t) \leq 1,$$

we have

$$0 \leq 1 - \tanh^2(t) \leq 1.$$

Therefore,

$$\boxed{\tanh'(t) \leq 1.}$$

—

(c) Gradient Bound for a Depth-L Network
For each hidden layer,

$$h_l = \phi(W_l h_{l-1}).$$

Using the chain rule,

$$\frac{\partial L}{\partial h_1} = \left(\prod_{l=2}^{L} D_l W_l \right) \frac{\partial L}{\partial h_L},$$

where

$$D_l = \operatorname{diag}\big(\phi'(z_l)\big)\,.$$

Taking spectral norms,

$$\left\| \frac{\partial L}{\partial h_1} \right\| \leq \prod_{l=2}^{L} \|D_l\|_2 \, \|W_l\|_2 \left\| \frac{\partial L}{\partial h_L} \right\|.$$

Since

$$\|D_l\|_2 \leq \max_t |\phi'(t)|,$$

we obtain

$$\boxed{\left\| \frac{\partial L}{\partial h_1} \right\| \leq \Big(\max_t |\phi'(t)| \Big)^{L-1} \prod_{l=2}^{L} \|W_l\|_2 \left\| \frac{\partial L}{\partial h_L} \right\|.}$$

—

(d) Sigmoid Case
For the sigmoid activation,

$$\max_t |\sigma'(t)| = \frac{1}{4}.$$

Hence,

$$\left\| \frac{\partial L}{\partial h_1} \right\| \leq \left(\frac{1}{4} \right)^{L-1} \prod_{l=2}^{L} \|W_l\|_2 \left\| \frac{\partial L}{\partial h_L} \right\|.$$

To prevent the gradient from vanishing, the weight matrices would need

$$\boxed{\|W_l\|_2 \approx 4}$$

for every layer so that the factor

$$\left(\frac{1}{4}\right)^{L-1} \prod_{l=2}^{L} \|W_l\|_2$$

remains close to one.

However, choosing very large spectral norms causes

- very large forward activations, - saturation of the sigmoid near 0 or 1, - unstable optimization and possible exploding activations or gradients.

Therefore,

Increasing the spectral norms to avoid vanishing gradients eventually causes instability and saturation.

---

(3) For a ReLU layer $z_l = W_l h_{l-1}$, $h_l = \mathrm{ReLU}(z_l)$, with i.i.d. zero-mean weights of variance v, derive He initialization. Repeat the derivation for (i) Leaky ReLU with slope a, and (ii) the absolute-value activation $\phi(t) = |t|$. Explain why the result for $|t|$ coincides with that of a linear network.

Solution.

(a) ReLU Activation

For a neuron,

$$z_l = W_l h_{l-1}.$$

Assume

- weights are independent with

$$E[W] = 0, \qquad \mathrm{Var}(W) = v,$$

- each neuron has fan-in n.

Then

$$\mathrm{Var}(z_l) = nv \ \mathrm{Var}(h_{l-1}).$$

Since

$$h_l = \max(0, z_l),$$

and z_l is symmetric about zero,

$$P(z_l > 0) = \frac{1}{2}.$$

Hence

$$E[h_l^2] = \frac{1}{2} E[z_l^2].$$

Therefore,

$$\mathrm{Var}(h_l) = \frac{1}{2} \mathrm{Var}(z_l).$$

Substituting,

$$\mathrm{Var}(z_{l+1}) = nv \cdot \frac{1}{2} \mathrm{Var}(z_l).$$

To keep the variance constant across layers,

$$nv \cdot \frac{1}{2} = 1.$$

Hence,

$$\boxed{v = \frac{2}{n}.}$$

This is the He initialization.

—

(b) Leaky ReLU

For Leaky ReLU,

$$\phi(z) = \begin{cases} z, & z > 0, \\ az, & z < 0, \end{cases}$$

where $0 < a < 1$.

Again,

$$P(z > 0) = P(z < 0) = \frac{1}{2}.$$

Thus,

$$E[\phi(z)^2] = \frac{1}{2}E[z^2] + \frac{1}{2}a^2E[z^2].$$

Therefore,

$$\mathrm{Var}(h_l) = \frac{1 + a^2}{2}\,\mathrm{Var}(z_l).$$

Hence,

$$\mathrm{Var}(z_{l+1}) = nv\frac{1 + a^2}{2}\,\mathrm{Var}(z_l).$$

Keeping the variance unchanged gives

$$nv\frac{1 + a^2}{2} = 1.$$

Therefore,

$$\boxed{v = \frac{2}{n(1 + a^2)}.}$$

—

(c) Absolute-Value Activation

For

$$\phi(z) = |z|,$$

we have

$$|z|^2 = z^2.$$

Hence,

$$E[\phi(z)^2] = E[z^2].$$

Therefore,

$$\mathrm{Var}(h_l) = \mathrm{Var}(z_l).$$

Thus,

$$\mathrm{Var}(z_{l+1}) = nv\ \mathrm{Var}(z_l).$$

To preserve variance,

$$nv = 1.$$

Hence,

$$\boxed{v = \frac{1}{n}.}$$

—

(d) Why $|t|$ Has the Same Initialization as a Linear Network
Although

$$|t|$$

is nonlinear,

$$|t|^2 = t^2.$$

Thus, the second moment of the activation is unchanged.
Since variance propagation depends only on

$$E[h^2],$$

the absolute-value activation preserves variance exactly like the identity function.
Therefore,

$$\boxed{\text{The initialization for } |t| \text{ is identical to that of a linear network: } v = \frac{1}{n}.}$$

---

(4) Let Swish be $s_\beta(t) = t\sigma(\beta t)$. Show that $s_\beta \to t/2$ as $\beta \to 0$ and $s_\beta \to \mathrm{ReLU}(t)$ as $\beta \to \infty$. For $\beta = 1$, show that s is non-monotonic with a unique global minimiser $t^* < 0$ satisfying

$$1 + t^*(1 - \sigma(t^*)) = 0.$$

Prove the identity $s(t^*) = t^* + 1$, then compute t^* to four decimal places using Newton's method, showing three iterations from $t_0 = -1$.

Solution.

(a) Limits of the Swish Function
The Swish activation is

$$s_\beta(t) = t\sigma(\beta t),$$

where

$$\sigma(z) = \frac{1}{1 + e^{-z}}.$$

As

$$\beta \to 0,$$

we have

$$\sigma(\beta t) \to \frac{1}{2}.$$

Therefore,

$$\boxed{\lim_{\beta \to 0} s_\beta(t) = \frac{t}{2}.}$$

—

As

$$\beta \to \infty,$$

$$\sigma(\beta t) \to \begin{cases} 1, & t > 0, \\ 0, & t < 0. \end{cases}$$

Hence,

$$\boxed{\lim_{\beta \to \infty} s_\beta(t) = \max(0, t) = \mathrm{ReLU}(t).}$$

—

(b) Global Minimum for $\beta = 1$
For

$$s(t) = t\sigma(t),$$

differentiate using the product rule,

$$s'(t) = \sigma(t) + t\sigma'(t).$$

Since

$$\sigma'(t) = \sigma(t)(1 - \sigma(t)),$$

we obtain

$$s'(t) = \sigma(t)\left[1 + t(1 - \sigma(t))\right].$$

Since

$$\sigma(t) > 0,$$

the stationary point satisfies

$$\boxed{1 + t^*(1 - \sigma(t^*)) = 0.}$$

There is only one solution with

$$t^* < 0,$$

so Swish has one unique global minimum.

—

(c) Identity $s(t^*) = t^* + 1$

From

$$1 + t^*(1 - \sigma(t^*)) = 0,$$

we obtain

$$1 + t^* = t^*\sigma(t^*).$$

Since

$$s(t) = t\sigma(t),$$

it follows that

$$\boxed{s(t^*) = t^* + 1.}$$

—

(d) Newton's Method

Define

$$f(t) = 1 + t(1 - \sigma(t)).$$

Then

$$f'(t) = 1 - \sigma(t) - t\sigma'(t).$$

Newton's iteration is

$$t_{k+1} = t_k - \frac{f(t_k)}{f'(t_k)}.$$

Starting with

$$t_0 = -1,$$

the iterations are

$$t_1 = -1.2635,$$
$$t_2 = -1.2783,$$
$$t_3 = -1.2785.$$

Thus,

$$\boxed{t^* \approx -1.2785.}$$

Using the identity,

$$\boxed{s(t^*) = t^* + 1 \approx -0.2785.}$$